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Practical 1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Execution task:

```
jamuna@Jem:~$ nano practical1.c
jamuna@Jem:~$ gcc practical1.c
jamuna@Jem:~$ ./a.out
Enter a command:
ls

Parent PID:923
Child PID:932

Child Process completed.

Child PID:932
Parent PID:815
a      ac      dir5      execl1.c  f1.c  fork.c      linux_task4  permission_lab  practical1.c
a.c    b1      dir6      execlp.c  f2    fork2.c     newfile1     pipe.c          read1.c
a.out  b2      execcv.c  execlp1.c f3    fork2.c.save open2.c       pipe2.c         readwrite.c
a7     content.c execcv.c.save execv.c   f4    fork3.c     open2.c.save pipex.c         student_portal
a8     d        execl.c   f1        fi    fork3.c.save pen2.c        pipex2.c        tt
```

```
jamuna@Jem:~$ uname
Linux
```

```

jamuna@Jem:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                   6
Model:                        186
Thread(s) per core:           2
Core(s) per socket:           6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4992.00
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
ear serialize ibt flush_lid arch_capabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           288 KiB (6 instances)
L1i:                           192 KiB (6 instances)
L2:                             7.5 MiB (6 instances)
L3:                             12 MiB (1 instance)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):            0-11
Vulnerabilities:
Gather data sampling:          Not affected
Ghostwrite:                    Not affected
Indirect target selection:     Not affected
Itlb multihit:                 Not affected
L1tf:                           Not affected
Mds:                           Not affected
Meltdown:                      Not affected
Mmio stale data:               Not affected
Old microcode:                 Not affected
Reg file data sampling:        Mitigation; Clear Register File
Retbleed:                      Mitigation; Enhanced IBRS
Spec rstack overflow:          Not affected
Spec store bypass:             Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                    Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                    Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_D
IS_S
Srbds:                          Not affected
Tsa:                           Not affected
Tsx async abort:               Not affected
Vmscape:                       Not affected

```

```

jamuna@Jem:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro
/

```

```

jamuna@Jem:~$ ps
  PID TTY          TIME CMD
  816 pts/0        00:00:00 bash
  997 pts/0        00:00:00 ps

```

```

jamuna@Jem:~$ top
top - 03:45:43 up 19 min, 1 user, load average: 0.05, 0.04, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7762.6 total, 7126.9 free, 545.9 used, 240.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7216.7 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24140	15388	11604	S	0.0	0.2	0:00.97	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
49	root	19	-1	50400	16784	15504	S	0.0	0.2	0:00.22	systemd-journal
81	systemd+	20	0	22416	14484	11984	S	0.0	0.2	0:00.08	systemd-resolve
87	root	20	0	35364	12328	9240	S	0.0	0.2	0:01.17	systemd-udev
107	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.07	chronyd-starter
108	root	20	0	4472	3140	2892	S	0.0	0.0	0:00.01	cron
109	message+	20	0	8820	5452	4772	S	0.0	0.1	0:00.08	dbus-daemon
115	root	20	0	44096	29892	14752	S	0.0	0.4	0:00.31	networkd-dispat
150	root	20	0	18436	9412	8168	S	0.0	0.1	0:00.09	systemd-logind
177	syslog	20	0	220548	5936	4660	S	0.0	0.1	0:00.16	rsyslogd
218	root	20	0	5492	2776	2588	S	0.0	0.0	0:00.01	agetty
232	_chrony	20	0	23684	11196	7224	S	0.0	0.1	0:00.11	chronyd
240	_chrony	20	0	12096	2436	1796	S	0.0	0.0	0:00.00	chronyd
273	root	20	0	123460	32464	17856	S	0.0	0.4	0:00.18	unattended-upgr
334	root	20	0	8648	5584	4740	S	0.0	0.1	0:00.02	login
379	jamuna	20	0	22336	13512	10888	S	0.0	0.2	0:00.24	systemd
381	jamuna	20	0	22912	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
411	jamuna	20	0	6136	5436	3864	S	0.0	0.1	0:00.07	bash
814	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
815	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.04	Relay(816)
816	jamuna	20	0	6268	5600	3928	S	0.0	0.1	0:00.13	bash
1007	jamuna	20	0	8180	6016	3900	R	0.0	0.1	0:00.00	top